

UNIVERSITY OF ESSEX ONLINE

MASTER OF SCIENCE THESIS

Measuring Technical Debt in Mission Critical Trading Systems

A Service Level Quantitative Framework

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Computing Department



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Declaration of Authorship

I, Tobias Zeier, declare that this thesis titled *Measuring Technical Debt in Mission Critical Trading Systems*, and the work presented in it are my own. I confirm that:

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- Where any part of this thesis has previously been submitted for a degree or any other qualification at this University or any other institution, this has been clearly stated.
- Where I have consulted the published work of others, this is always clearly attributed.
- Where I have quoted from the work of others, the source is always given. With the exception of such quotations, this thesis is entirely my own work.
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Abstract

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Master of Science

Measuring Technical Debt in Mission Critical Trading Systems

by Tobias Zeier – 12696372

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List of Abbreviations

AHP	Analytic Hierarchy Process
BCS	British Computer Society
CFR	Change Failure Rate
CI/CD	Continuous Integration / Continuous Delivery
CMDB	Configuration Management Database
COTS	Commercial Off the Shelf
DORA	DevOps Research and Assessment
DSR	Design Science Research
FINMA	Swiss Financial Market Supervisory Authority
FADP	Federal Act on Data Protection
IT	Information Technology
ITIL	IT Infrastructure Library
ITSM	IT Service Management
MCDM	Multi-Criteria Decision-Making
MTTR	Mean Time to Restore
P1-3	Priority 1-3
RQ	Research Question
SAW	Simple Additive Weighting
SLA	Service Level Agreement
SQALE	Software Quality Assessment based on Lifecycle Expectations
TD	Technical Debt
TOPSIS	Technique for Order of Preference by Similarity to Ideal Solution
VIKOR	Vlsekriterijumska Optimizacija I Kompromisno Resenje

List of Symbols

C_i	TOPSIS closeness coefficient
S_i^+	Euclidean distance from positive ideal solution
S_i^-	Euclidean distance from negative ideal solution
x_{ij}	Raw indicator value for service i , criterion j
r_{ij}	Vector-normalised indicator value
v_{ij}	Weighted normalised indicator value
w_j	Criterion weight
m	Number of services evaluated
n	Number of indicators
τ	Kendall's tau, a ranking correlation coefficient

1 Introduction

Technical debt is not an abstract software quality concern; in mission-critical trading IT it translates into delayed recovery, accumulating infrastructure risk and governance blind spots. The following sections establish why the absence of a quantitative, service-level measurement mechanism represents a specific and costly gap, and how this thesis addresses it.

1.1 Background and Motivation

Technical debt describes the implied future cost of accepting short-term technical compromises in place of more robust solutions (Cunningham, 1992). Since its introduction, the concept has expanded from source code quality to encompass architectural, design, test and documentation dimensions of complex IT systems (Kruchten, Nord and Ozkaya, 2012). Its operational and financial consequences are measurable and compound over a small number of years, as Chapter 2 examines in detail (Banker, Liang and Ramasubbu, 2020). In financial trading environments, these consequences are intensified by continuous availability requirements, vendor-dominated architectures and regulatory oversight. This makes the absence of a quantitative prioritisation mechanism particularly costly (Ramasubbu and Kemerer, 2021; Haki *et al.*, 2023).

1.2 Problem Statement

The technical debt literature lacks a standardised, quantitative framework for measuring and prioritising debt at the service level in vendor-dominated enterprise IT. Lenarduzzi *et al.* (2021) reviewed 44 primary studies on prioritisation and found that quantitative multi-criteria approaches are uncommon and empirical validation on industrial datasets is rare. Dominant measurement tools

operationalise debt as remediation effort derived from source code rule violations (Biazotto *et al.*, 2024), rendering them inapplicable where source code access is unavailable, as is typical in trading IT. Even where code is accessible, tools applied to the same codebase produce substantially divergent estimates (Amanatidis *et al.*, 2020), leaving infrastructure lifecycle debt, platform obsolescence and vulnerability debt largely unaddressed. Although DORA metrics have been validated as predictors of software delivery performance (Forsgren, Humble and Kim, 2018), the inferential step from degraded operational indicators to elevated technical debt severity is not yet empirically established and is treated in this study as a testable hypothesis. None of the literature reviewed in the context of this thesis applied a multi-criteria decision-making framework to live operational service management data in order to produce a ranked technical debt prioritisation instrument for financial trading IT. This specific absence motivates the framework developed in this thesis.

1.3 Research Aims and Objectives

This study develops a service-level quantitative framework for measuring and prioritising technical debt in mission-critical trading systems, grounded in operational service management data rather than source code analysis. The specific objectives are as follows.

1. To review literature on technical debt measurement, prioritisation, financial enterprise IT governance and multi-criteria decision-making.
2. To identify and operationally define quantifiable technical debt indicators derived from live ITSM and CMDB records.
3. To design and apply a TOPSIS-based multi-criteria model for scoring and ranking services by technical debt priority.
4. To evaluate the framework by testing ranking stability under weight and temporal variation, and by assessing whether the resulting rankings are coherent with independent operational evidence.

1.4 Research Questions

The study is guided by the following primary research question.

RQ1: How can technical debt in mission-critical trading systems be prioritised quantitatively at the service level using operational data?

Two secondary questions support the investigation.

- RQ2: To what extent do operational service management metrics serve as valid proxies for technical debt severity in vendor-dominated enterprise IT environments?
- RQ3: How stable are the resulting service rankings when criterion weights are perturbed?

1.5 Significance of the Study

The main academic contribution is a working prioritisation framework that did not previously exist for this type of environment, connecting how services are managed operationally with how technical debt is typically studied in the software engineering literature. To the best of current knowledge, no prior study has applied TOPSIS to live operational records from a financial trading institution for technical debt prioritisation. Practically, the framework operates on data routinely available in enterprise ITSM and CMDB systems without requiring source code access, making it directly applicable to vendor-dominated trading IT architectures and supporting auditable remediation decisions in regulated environments (*FINMA Circular 2023/1 - Operational risks and resilience – banks, 2024*).

The framework measures debt indirectly, through operational proxies, and expresses the result as a relative priority rather than an absolute quantity. The word “measuring” in the title should be read in that sense: a quantitative, reproducible ordering of services by debt priority, not a claim to have quantified the debt itself.

1.6 Thesis Structure

Chapter 2 reviews the theoretical and empirical foundations across four bodies of literature and identifies the research gap. Chapter 3 describes the research design, methodology and ethical considerations. Chapter 4 presents the framework development and results, including TOPSIS computation and sensitivity analysis. Chapter 5 evaluates the framework against the criteria established in Chapter 3. Chapter 6 concludes with responses to the research questions, a critical self-evaluation, the academic and practical contributions, and directions for future work.

2 Literature Review

Four bodies of literature intersect at the problem this thesis addresses: the theory of technical debt, approaches to its measurement and prioritisation, its governance in regulated financial IT, and multi-criteria decision-making. None of these streams alone is sufficient. Read together and critically, they expose a gap that the proposed framework is designed to fill.

2.1 Context of the Topic

Technical debt has become both a central concept and a persistent governance problem in software engineering and enterprise IT (Avgeriou *et al.*, 2023). Ward Cunningham introduced the metaphor in 1992, describing technical debt as the long-term cost of short-term technical compromises (Cunningham, 1992). Over time, the concept expanded from source code quality to encompass architectural, infrastructure, organisational and process dimensions of complex IT systems (Kruchten, Nord and Ozkaya, 2012).

Despite this conceptual broadening, the empirical and tool-supported literature remains concentrated at the code level. Static analysis tools and code-based metrics work tolerably well for greenfield development dominated by in-house code but are much less appropriate for enterprise environments where reliability depends on vendor platforms, legacy infrastructure and tightly integrated service stacks (Amanatidis *et al.*, 2020). In such environments, forms of technical debt cannot be detected via code-centric tools.

What the literature does not yet offer is a way to measure and rank technical debt in environments where source code is unavailable. That absence is what this review is trying to account for. No study reviewed demonstrated a framework that provides a quantitative, service-level mechanism for measuring and

prioritising technical debt in environments such as trading IT. In these contexts, significant forms of technical debt manifest not only in code repositories but also in operational outcomes such as service instability, patch delay and infrastructure obsolescence (Ramasubbu and Kemerer, 2021; Haki *et al.*, 2023). The following sections build a theoretical and methodological basis for the proposed framework.

2.2 Methodology for Literature Selection

The literature search was conducted between January and April 2026 using a systematic and reproducible approach, drawing on four principal academic databases: the University of Essex library, Scopus, IEEE Xplore and Google Scholar. Search terms are summarised in Table 2.1.

TABLE 2.1: Search terms used for literature review

Theme	Search String
TD foundations	"technical debt" AND (metaphor OR taxonomy OR typology OR theory)
TD measurement and tools	"technical debt" AND (measurement OR "static analysis" OR SonarQube OR metric OR "code smell")
TD prioritisation	"technical debt" AND (priorit* OR "backlog management" OR remediation OR repayment)
TD in enterprise and financial IT	"technical debt" AND ("enterprise" OR "financial services" OR "trading" OR "legacy system" OR "regulated")
TD automation	"technical debt management" AND ("machine learning" OR intelligent OR automation OR NLP)
MCDM and TOPSIS	TOPSIS AND ("multi-criteria" OR MCDM OR "SAW" OR "decision making" OR priorit*)
Patch management	"patch management" AND (priorit* OR risk OR vulnerability OR SLA OR enterprise)

Theme	Search String
Operational risk and DORA	("DORA metrics" OR "mean time to restore") AND ("technical debt" OR "software delivery")

Sources were screened against the pre-specified inclusion and exclusion criteria in Table 2.2.

TABLE 2.2: Inclusion and exclusion criteria

Criterion	Rule
Publication type	Peer-reviewed journal articles, conference proceedings or empirical technical reports
Language	English full text available
Date range	2019–2026, or foundational regardless of date
Thematic relevance	Directly addresses at least one of the four thematic pillars
Exclusion: scope	Opinion pieces without empirical grounding
Exclusion: topic	Agile process improvement without a measurable TD component
Exclusion: duplicates	Superseded by a more comprehensive journal version

Screening was conducted in two stages. Titles and abstracts were first assessed against these criteria, reducing the initial pool. Remaining sources underwent full-text review to verify relevance and methodological quality. Forward and backward citation snowballing was subsequently applied to three anchor publications: Kruchten, Nord and Ozkaya (2012), Lenarduzzi *et al.* (2021) and Forsgren, Humble and Kim (2018) to surface relevant studies not captured by database searches. The process yielded approximately 110 candidate sources, of which more than 50 were retained following full-text screening.

The review uses narrative synthesis rather than a formal systematic review protocol. The research spans three disciplines, software engineering, enterprise IT governance and decision science, which differ significantly in their research

methods, terminology and evidence standards. A single meta-analytic framework would not accommodate this heterogeneity, and the aim of the review is to build a coherent argument in support of a novel design science artefact, for which narrative synthesis is the appropriate approach (Hevner *et al.*, 2004). The following four sections build the theoretical and contextual case for the proposed framework in sequence.

2.3 Conceptual Foundations of Technical Debt

Before examining how technical debt is measured, it is worth understanding how the concept has evolved. Most of the tools reviewed in the next section are still calibrated to an earlier, narrower version of technical debt, and that mismatch is part of the problem this thesis is trying to address.

2.3.1 Origins and the Metaphor

Cunningham (1992) introduced the metaphor to explain a refactoring strategy to non-technical stakeholders. In his original formulation, shipping code that is “not quite right” resembles taking on financial debt, each subsequent maintenance effort represents interest. If the principal is not repaid through refactoring, the cost of working with the system compounds until change becomes untenable.

The metaphor has proved productive but carries inherent limits. In financial debt, the principal is contractually defined, and the interest rate is known in advance. In technical debt, neither holds: the principal is an estimated remediation cost that changes as systems evolve, and the interest depends on development activity that is uncertain at the time of incurrence. These limitations matter directly for this thesis. The proposed framework does not rely on the metaphor for rigour; it grounds its estimates in service-level operational data rather than in abstract financial analogies, making interest costs concrete and operationally observable.

Several influential practitioners refined the concept. McConnell (2008) introduced a cost structure distinguishing principal, recurring interest and compounding interest, and differentiated unintentional debt from intentional debt incurred as a conscious trade-off. Fowler (2009) proposed a quadrant distinguishing

reckless from prudent debt and deliberate from inadvertent debt. The quadrant originates in a practitioner blog post rather than a peer-reviewed publication, and its value lies in stakeholder communication rather than theoretical grounding. Kruchten, Nord and Ozkaya (2012) subsequently formalised a taxonomy covering code, design, architectural, test, build and documentation debt on the basis of a structured research programme, and this taxonomy underpins much of the subsequent empirical work.

2.3.2 Theoretical Development

Empirical studies have confirmed and extended this view. Besker (2020) reported that developers in industrial settings spend roughly 23% of their time dealing with the consequences of existing technical debt, with architectural debt exerting the strongest negative influence on productivity. Bedi and Kaur (2022) used regression analysis on open-source repositories and found that commit frequency, lines of code, code smells, and test coverage are significant predictors of technical debt, whereas developer reputation metrics are not. This suggests that technical debt is driven primarily by development practices rather than by the perceived quality of individual contributors.

Ahmad *et al.* (2026) approached the topic qualitatively, interviewing experts from four international companies. They identified five non-technical dimensions of debt accumulation in large-scale agile environments, namely social dynamics, process, people, documentation and requirements, with lack of communication emerging as the dominant driver. This reinforces the view of technical debt as a sociotechnical phenomenon that cannot be governed by technical measurement alone.

Vidoni, Codabux and Fard (2022) advanced a more radical argument through a game-theoretic lens, proposing that in complex systems technical debt cannot be completely removed and that its effects cannot be fully controlled even when individual items are repaid. The concept of “infinite technical debt” challenges the assumption, implicit in many repayment-focused frameworks, that debt is always recoverable given sufficient effort. This argument has not yet been empirically validated in enterprise IT settings and should be understood as a

theoretical provocation rather than an established finding. Governance outputs should therefore surface replacement decisions where service-level data suggests debt has crossed a threshold beyond which incremental repayment is unlikely to be effective.

2.3.3 Expanding the Conceptual Perimeter

The conceptual perimeter of technical debt has progressively widened beyond source code to encompass systems engineering, regulatory lifecycle and vendor platform concerns. Kleinwaks, Batchelor and Bradley (2023) found in a systematic review that requirements, architecture, integration artefacts and verification artefacts can all accumulate debt, and that software-oriented approaches are poorly adapted to multidisciplinary systems engineering lifecycles.

Hanssen, Brataas and Martini (2019) identified a particularly important mechanism: regulatory change as a debt trigger. In an open banking case study, architectural decisions that were adequate before the EU Payment Services Directive 2 (PSD2) became sources of scalability debt once new requirements were imposed without architectural review. This illustrates how external regulatory change can convert previously adequate designs into operational liabilities. This pattern is directly relevant to financial trading IT, where regulatory obligations evolve continuously, and new compliance requirements can render existing architectures inadequate overnight.

Yang *et al.* (2019) proposed a taxonomy of Commercial Off-the-Shelf (COTS) related technical debt covering seven key types, including vendor platform obsolescence and integration constraints. For financial trading IT, where vendor platforms and COTS products form the core of the operational stack, this broader view is essential: platform debt, infrastructure debt and vulnerability debt are not detectable through standard static analysis. Managing them requires indicators derived from infrastructure records and operational monitoring rather than from code repositories.

2.4 Measurement and Prioritisation Frameworks

Measurement is where the gap between the concept and current practice becomes most visible. The dominant tools operationalise technical debt as a function of source code, which works tolerably well in greenfield development but leaves infrastructure debt, platform debt and vulnerability debt largely invisible.

2.4.1 The Dominance of Static Analysis and Its Limits

Static code analysis tools such as SonarQube, SQALE and CAST dominate technical debt measurement, operationalising debt as estimated remediation effort derived from rule violations (Biazotto *et al.*, 2024). In a systematic mapping of 178 primary studies, Biazotto *et al.* (2024) found that most automation artefacts remain code-level, require substantial human intervention even at higher automation levels, and rarely support end-to-end technical debt management. The focus on code leaves infrastructure-level, platform-level and vendor-related debt only weakly observed.

This limitation is material in enterprise environments where operational reliability depends on legacy infrastructure and third-party platforms rather than on in-house code alone. Ampatzoglou *et al.* (2022) presented SDK4ED as a comprehensive industry-oriented platform integrating several code-level techniques, reporting financial benefits over a three-year evaluation. Even so, its support for infrastructure and vendor-platform debt remains limited. Tornhill and Borg (2022) confirmed the business relevance of code-level debt across 39 proprietary codebases, demonstrating that poor code quality was statistically associated with higher defect density, slower remediation and lower developer throughput. However, this does not resolve the blind spots created by a code-only perspective. Organisations that concentrate remediation resources on refactoring may simultaneously leave critical legacy infrastructure and unsupported components unaddressed.

A further concern is measurement consistency. Amanatidis *et al.* (2020) found substantial divergence among technical debt assessment tools applied to the same 50 open-source projects: different tools produced different estimates and

highlighted different violation sets, with no single tool consistently dominating. A debt figure generated by any one tool therefore reflects that tool's ruleset and thresholds rather than an objective underlying quantity. Albuquerque *et al.* (2023) confirmed that infrastructure debt and process debt remain significantly understudied relative to code and architectural debt. Taken together, these findings support the view that static analysis is informative but insufficient for service-level prioritisation in vendor-dominated enterprise IT.

2.4.2 The Prioritisation Gap

Lenarduzzi *et al.* (2021) analysed 44 primary studies on technical debt prioritisation, finding that existing approaches cluster around cost-based, value-based and risk-based strategies, that most adopt a single primary criterion, that quantitative methods are less common than heuristic techniques, and that empirical validation on industrial datasets is rare. The field lacks agreement on which factors matter, how to measure them and what constitutes an adequate prioritisation process. This conclusion directly motivates the quantitative contribution of the present work.

The operational stakes of inadequate prioritisation are concrete. de Toledo *et al.* (2021) showed in a production microservices system that targeted repayment of architectural technical debt produced an 84% reduction in incidents, grounding the case for systematic prioritisation in a measurable outcome rather than in theoretical argument. Empirical studies consistently show that practitioners consider both value and cost rather than a single strategy, and that perceived value is highly context-dependent (Alfayez *et al.*, 2023). Pina, Seaman and Goldman (2022) found that prioritisation decisions reflect a combination of technical severity, business impact and available effort, and that developers frequently experience tension between technically optimal choices and organisationally imposed constraints. Most IT managers nonetheless lack a formal technical debt management process, with cross-functional communication identified as the primary obstacle (Wiese and Borowa, 2023).

Stochel *et al.* (2023) proposed the Continuous Debt Valuation Approach (CoDVA), assigning a financial value to each debt item based on remediation

cost, interest rate and business risk exposure. Case studies show that financial valuation can significantly change repayment priorities relative to effort-only approaches. However, CoDVA still relies on development-level data and does not incorporate operational service management signals such as incident rates or change failure rates. These are precisely the signals available in vendor-dominated environments where source code is inaccessible. These studies collectively establish that any quantitative prioritisation framework must make its value judgements transparent and auditable, particularly in regulated institutions where priority scores may influence funding, staffing and post-incident accountability.

2.5 Technical Debt in Financial and Regulated Enterprise IT

The limitations identified in the previous section are compounded in financial services. Vendor-managed architectures, regulatory obligations and continuous availability requirements create a context that the existing measurement literature was not designed for, and understanding that context is essential before proposing a framework intended to operate within it.

2.5.1 The Financial Services Context

Financial services organisations operate under continuous availability obligations, layered regulatory compliance requirements and long-lived vendor-managed architectures that cannot be freely refactored. These conditions shape both how technical debt accumulates and how it must be measured, and they define the context within which the studies reviewed in this section are read.

The most directly relevant empirical grounding for the present framework comes from Haki *et al.* (2023), who conducted clinical research at Credit Suisse across more than 3,000 IT applications and defined a technical debt taxonomy covering code quality debt, infrastructure lifecycle debt, vulnerability debt and automation debt. Infrastructure lifecycle debt was measured as cumulative months of unsupported components; vulnerability debt as weighted scanner-detected

vulnerability counts. This taxonomy matters for a specific reason: it demonstrates that a major regulated financial institution treats infrastructure lifecycle debt and vulnerability debt as operationally significant categories alongside code-level debt, and measures both through infrastructure records rather than source code analysis. That precedent directly supports the broader measurement approach taken in this thesis. The conceptual contribution stands on its own; the empirical scope, however, does not support generalisation beyond that institution's specific context, as the quantitative evaluation covered only three development teams within a single organisation.

Rolland and Lyytinen (2021) investigated architectural debt accumulation at a large Scandinavian financial institution, referred to in the study by the pseudonym BankAlpha, through eleven in-depth interviews. Two tensions emerged consistently: decentralised feature teams optimising for local throughput against the governance needed to manage debt at system level, and the pace of digital innovation against the long-term sustainability of the supporting architecture. Both are intensified in financial services by compliance obligations and competitive pressure to release products rapidly. The finding reinforces a pattern also visible in Haki *et al.* (2023): organisational dynamics drive debt accumulation at least as much as technical decisions do, and management intent is the decisive governance variable in both cases.

2.5.2 Enterprise Systems and the Outsourcing Context

Banker, Liang and Ramasubbu (2020) examined 26 firms operating a COTS CRM system over its full eleven-year lifecycle. Using propensity score matching against firms with zero technical debt, they found that a 10% increase in technical debt reduced gross return on assets by 16% on average. Critically, this penalty more than doubled within six years of adoption as code decay compounded structural complexity. While all firms initially benefited from system customisation, those carrying median debt levels saw the net value of their system turn negative by year five, a pattern robust to alternative performance measures including asset turnover, gross margin and operating income. This shows that technical debt in vendor-managed platforms produces measurable financial harm that

compounds over a relatively short horizon, with the net value of a median-debt system turning negative within roughly five years.

Ramasubbu and Kemerer (2021) extended this picture to 1,824 outsourced remediation projects, finding that client-vendor coordination only improved outcomes when both parties had agreed a long-term migration roadmap. Without this, increased coordination actually degraded performance. Rinta-Kahila *et al.* (2023) showed through system dynamics modelling how COTS customisation creates self-reinforcing debt accumulation loops that code-level tools cannot detect. Both findings support the governance argument of the present work. Ranking debt items must be embedded in a strategic remediation context, and governance-level visibility of infrastructure debt is required before those loops become irreversible.

2.5.3 DORA Metrics as Operational Debt Proxies

Forsgren, Humble and Kim (2018) drew on survey responses from over 2,000 unique organisations worldwide and showed that DORA metrics MTTR, CFR, deployment frequency and lead time can distinguish high-performing from low-performing delivery teams. This makes these indicators useful in a vendor-dominated environment where source code inspection is not available.

The inferential step this thesis takes, however, goes beyond what the original validation supports. Forsgren, Humble and Kim (2018) established DORA metrics as delivery performance predictors. Using them as proxies for technical debt severity requires a further claim. Degraded operational performance reflects accumulated debt rather than inadequate staffing, immature release practices or organisational disruption. That claim is plausible but has not been established as a general empirical law. The framework therefore treats this as a hypothesis rather than a premise. The literature does not settle the question, and it would be misleading to present it as if it did.

Nord, Ozkaya and Woody (2021) provide the most direct grounding for this inferential step. Drawing on practitioner examples and software engineering literature, they classify delivery tempo degradation and deployment instability as observable technical debt symptoms that manifest in production records without requiring source code access. Critically, they identify elevated MTTR and CFR

not as general performance shortfalls but as indicators that accumulated debt is actively impeding the system's ability to absorb and recover from change. That distinction matters for the framework developed in this study. It positions MTTR and CFR as diagnostic signals of debt severity. Paudel *et al.* (2025) found that technical debt density explained between 5% and 41% of lead time variance across six fintech components, with team structure and component ownership acting as confounding factors. This reinforces the case for multi-criteria measurement rather than reliance on any single proxy.

Nielsen and Skaarup (2022) offer complementary evidence from a vendor-managed public sector portfolio. Operational errors emerged shortly after TD-related resource constraints were imposed on a shared service component, and unsupported component versions became visible only during infrastructure migration rather than through routine monitoring. Although the public sector context limits direct transferability, the operational mechanism is consistent with the vendor-dominated profile of the case organisation in the present study.

The following table maps each operational indicator to the technical debt dimension it proxies and the literature that justifies its inclusion.

TABLE 2.3: Mapping of operational indicators to technical debt dimensions and literature justification

Indicator	TD Dimension	Literature Justification
Incident frequency	Architectural debt	de Toledo <i>et al.</i> (2021); Ramasubbu and Kemerer (2021); Ramasubbu, Kemerer and Woodard (2015); Nielsen and Skaarup (2022)
Mean Time to Restore (MTTR)	Infrastructure / platform debt	Forsgren, Humble and Kim (2018); Nord, Ozkaya and Woody (2021); Paudel <i>et al.</i> (2025)
Change failure rate	Process debt	Forsgren, Humble and Kim (2018); Nord, Ozkaya and Woody (2021); Paudel <i>et al.</i> (2025)

Indicator	TD Dimension	Literature Justification
Patch recency	Vulnerability debt	Tizio, Armellini and Massacci (2023); Markkanen and Frantti (2023)
Unsupported component months	Infrastructure lifecycle debt	Haki <i>et al.</i> (2023)

2.5.4 Security Debt and Patch Management

Security-related technical debt is most visible in the literature on patch management and vulnerability handling. Tizio, Armellini and Massacci (2023) analysed 86 Advanced Persistent Threat across 350 campaigns from 2008 to 2020, finding that enterprises delaying updates by one month faced 4.9 times higher odds of compromise compared to immediate patching, rising to 9.1 times at three months. Contrary to common assumption, most campaigns exploited publicly known vulnerabilities rather than zero-days, confirming that unpatched systems carry measurable, quantifiable risk. These figures require careful contextualisation: the dataset spans all industries and predates regulatory changes, including FINMA circular 2023/1 and the EU DORA regulation, that have since accelerated patching cycles in financial services specifically (*FINMA Circular 2023/1 - Operational risks and resilience – banks*, 2024). The risk multipliers are therefore best treated as indicative of the direction and order of magnitude of the relationship rather than as directly applicable benchmarks for contemporary trading IT.

Markkanen and Frantti (2023) reviewed current NIST guidance and the academic literature on patch management planning. They argue that uniform, time-based patching policies are increasingly inadequate given the shrinking window between vulnerability disclosure and exploitation, and that organisations should treat patching as a risk-based, organisation-wide business priority rather than a purely technical responsibility. Nurse (2025) framed patch prioritisation as a problem of maintaining SLA compliance, noting that business partners and clients drive patching decisions through supply chain agreements and contractual requirements. This aligns patch management directly with the service-level

focus of this study.

Kula *et al.* (2019) studied dependency management at ING, showing that keeping upstream dependencies current is a persistent challenge in rapid-release banking environments and that delays contribute directly to technical debt. The study is based on a single organisation and may reflect ING-specific governance practices, so caution is warranted in applying its findings to trading IT environments with different release governance structures. Taken together, these studies establish that in financial trading IT, significant forms of technical debt extend well beyond source code: regulatory change can convert previously adequate architectures into liabilities (Hanssen, Brataas and Martini, 2019), unsupported components accumulate measurable infrastructure lifecycle debt (Haki *et al.*, 2023), and service instability signals accumulated debt that compounds with system growth (Ramasubbu and Kemerer, 2021). Each of these manifestations is detectable through ITSM and CMDB records rather than static analysis.

2.6 Multi-Criteria Decision-Making and TOPSIS

Prioritising technical debt involves balancing several criteria simultaneously: remediation cost, operational risk, security exposure and regulatory compliance. That is precisely the kind of problem multi-criteria decision-making methods are designed for, and this section makes the case for TOPSIS as the most appropriate choice among the available options.

A weighted scoring matrix is the simplest alternative, but it assumes that criteria are independent and commensurable across scales. Neither assumption holds reliably when combining continuous cost metrics with ordinal or binary compliance indicators. A risk matrix is categorical and does not support fine-grained ranking of a large service portfolio. TOPSIS addresses both limitations by normalising heterogeneous criteria before aggregation and ranking alternatives by their geometric proximity to the ideal solution, making it well suited to contexts where criteria differ in type, scale and direction of preference (Ciardiello and Genovese, 2023). Practitioners weigh these dimensions together rather than applying a single factor (Alfayez *et al.*, 2023), and Lenarduzzi *et al.* (2021) concluded that the

absence of a validated, quantitative, multi-criteria framework remains a central obstacle to evidence-based technical debt governance.

2.6.1 TOPSIS: Mechanics, Applications and Critical Limitations

Hwang and Yoon (1981) introduced TOPSIS, defining the ideal solution as the hypothetical alternative scoring best on all criteria simultaneously and ranking alternatives by their Euclidean proximity to that ideal. Zanakakis *et al.* (1998) showed that TOPSIS produces stable rankings close to the Pareto-optimal ordering across a range of problem structures, supporting its use in prioritisation tasks under uncertainty.

The procedure constructs a normalised weighted decision matrix, identifies ideal best and worst values per criterion, and ranks alternatives by their closeness coefficient, the ratio of the distance to the ideal worst to the sum of both distances (Albarak *et al.*, 2022). A well-documented limitation is rank reversal: adding or removing alternatives can shift the rankings of existing options (García-Cascales and Lamata, 2012). Yang (2020) proposed a normalisation adjustment that mitigates this by defining reference points independently of the current alternative set, which is particularly relevant for enterprise portfolios where services are regularly added or decommissioned.

Three alternative MCDM methods warrant comparison. VIKOR identifies compromise solutions between conflicting criteria rather than producing a global ranking, making it less appropriate when the required output is a ranked remediation list (Shekhovtsov and Sařabun, 2020). AHP derives weights through pairwise comparisons but does not natively produce a full portfolio ranking, and the comparison process grows burdensome as the number of criteria increases (Ishak and Wanli, 2020). A weighted scoring matrix is simpler still, but assumes criteria are measured on comparable scales, which does not hold when mixing continuous operational metrics with categorical compliance indicators. TOPSIS integrates normalisation, weighting and ranking into a single procedure, making it the most efficient choice for generating a ranked service portfolio. Where expert-derived weights are needed, AHP pairwise comparisons can serve as an

upstream step before feeding results into TOPSIS, a combination demonstrated by Ishak and Wanli (2020) in a regulated operational context.

Albarak *et al.* (2022) provide the most directly analogous precedent, combining TOPSIS with portfolio theory to prioritise technical debt and demonstrating that it surfaces high-priority items likely to be deferred under effort-only approaches. TOPSIS has also been applied in other risk-intensive, regulated contexts (Zytoon, 2020; Khan and Purohit, 2022). None of these studies, however, operates on live service management data in a large financial enterprise setting, which is a defining characteristic of this study.

2.7 Transferability and Data-Quality Critique

Although the preceding sections provide strong motivation for the proposed framework, none of the existing applications reviewed operates on live operational records in large enterprise settings. Most rely on researcher-defined scenarios, synthetic datasets or controlled development artefacts. The proposed research differs by applying TOPSIS to operational records such as incident logs, ITSM change data and vulnerability scan outputs, which are known to have quality issues. Incident classifications can vary between teams. Recorded change failure rates can be influenced by documentation practices as much as by technical quality. Vulnerability counts depend on scanner coverage and configuration. These data limitations are explicitly acknowledged throughout Chapter 4 and inform the interpretation of results in Chapter 5. The sensitivity of rankings to input variation is examined there through weight perturbation analysis, following the practice recommended for composite indicators by Greco *et al.* (2019), before any conclusions about priority ordering are drawn.

Using operational service management data as a basis for technical debt prioritisation also raises important ethical questions. Such records are generated and owned by operations teams, and development teams may not have consented to their use in performance evaluations. A framework that ranks services based on operational indicators may influence budgets, staffing and reputational assessments in regulated institutions. These considerations are addressed in full in the methodology chapter at Section 3.2.

2.8 Synthesis: Converging Evidence and the Research Gap

The four thematic sections together define the conceptual, methodological, contextual and technical landscape within which this research is situated.

2.8.1 Cross-Pillar Convergence

The four preceding sections converge on a common structural problem. The concept of technical debt has expanded beyond source code and now includes infrastructure, vendor platforms and operational dependencies, but the dominant measurement and prioritisation methods have not kept pace with that wider scope (Yang *et al.*, 2019; Kleinwaks, Batchelor and Bradley, 2023; Biazotto *et al.*, 2024). In regulated, vendor-heavy trading IT, that mismatch leaves important forms of debt only partially visible and difficult to govern effectively (Rinta-Kahila *et al.*, 2023).

The first implication is that the field now describes more forms of debt than its tools can reliably observe. Code-centric approaches remain useful, but they are poorly suited to environments where the most relevant liabilities sit in platform obsolescence, infrastructure lifecycle exposure or operational dependence on third-party systems (Haki *et al.*, 2023; Biazotto *et al.*, 2024). As a result, the literature supports the existence of these broader debt types while still offering limited practical means of measuring them in service-level terms (Kleinwaks, Batchelor and Bradley, 2023).

A second issue concerns prioritisation. Technical debt is inherently multi-criteria, because cost, value, risk and operational impact must all be considered together, yet the literature does not offer a validated quantitative method that combines these dimensions into a transparent service-level ranking (Lenarduzzi *et al.*, 2021; Pina, Seaman and Goldman, 2022; Alfayez *et al.*, 2023). In practice, this means that remediation decisions are often based on implicit judgement or single-factor reasoning, even though the problem itself is multidimensional.

A third strand of evidence concerns consequences. Unmanaged technical debt produces observable operational harm in the form of increased incidents, slower

recovery, lower productivity, higher remediation cost and greater vulnerability exposure (Banker, Liang and Ramasubbu, 2020; de Toledo *et al.*, 2021; Tornhill and Borg, 2022; Tizio, Armellini and Massacci, 2023). These effects are recorded in operational systems rather than source code repositories, which makes them visible in ITSM and CMDB data and therefore suitable for service-level analysis (Ramasubbu and Kemerer, 2021).

Taken together, these strands justify using operational indicators as proxies for prioritisation rather than as direct measures of technical debt itself (Haki *et al.*, 2023; Tizio, Armellini and Massacci, 2023). Their value lies in providing a practical, auditable basis for ranking services in environments where code-centric tools are unavailable or incomplete. This is the precise gap that the framework developed in this thesis is intended to address.

2.8.2 Cross-Cutting Limitations

Several limitations qualify these conclusions. The empirical base for financial trading IT specifically remains thin. Studies at Credit Suisse, at BankAlpha and at ING each provide valuable evidence but are restricted to single organisations (Kula *et al.*, 2019; Rolland and Lyytinen, 2021; Haki *et al.*, 2023). Whether the patterns observed generalise to other financial institutions, including securities trading environments with distinct operational profiles, remains an open question. This study contributes an operational dataset from a regulated financial trading environment that is currently absent from the technical debt literature.

Existing TOPSIS applications to technical debt rely on controlled datasets, and the comparative work of Shekhovtsov and Salabun (2020) shows that method choice matters less than weight specification when weights are uncertain. This underscores the importance of transparent and justified weight determination in the present study, which is addressed through a dual-configuration sensitivity analysis in Chapter 5. Behavioural and governance factors, including management intent, incentives and communication patterns, consistently exert a stronger influence on remediation outcomes than technical tools alone (Besker, 2020; Wiese and Borowa, 2023; Haki *et al.*, 2023). The scores the framework produces are a starting point for a governance conversation rather than a final answer. A

service ranked high priority still requires human judgement about whether and how to act.

2.8.3 Literature Coverage and Thesis Contribution

The following table maps the literature reviewed in this chapter to the thesis argument. For each thematic category, it identifies what existing sources cover, where the literature falls short, and precisely what this dissertation addresses in response. The table makes the novelty of the contribution explicit because no existing study combines all four elements, namely service level indicators, multi criteria prioritisation, live operational data, and a trading IT context, into a single auditable framework.

TABLE 2.4: Overview of related work and research gap

Reference	Focus	SL Ind.	Multi-crit.	Fin. IT	Key gap
(Kruchten, Nord and Ozkaya, 2012); (Fowler, 2009); (Cunningham, 1992)	TD taxonomy	×	×	×	No measurement mechanism
(Biazotto <i>et al.</i> , 2024); (Amanatidis <i>et al.</i> , 2020); (Ampatzoglou <i>et al.</i> , 2022)	Static analysis tools	×	×	×	Inapplicable to vendor platforms
(Lenarduzzi <i>et al.</i> , 2021); (Pina, Seaman and Goldman, 2022); (Alfayez <i>et al.</i> , 2023)	TD prioritisation	×	Partial	×	Single criterion; no operational data
(Stochel <i>et al.</i> , 2023); (Wiese and Borowa, 2023)	Business-driven valuation	×	×	×	No service-level signals
(Haki <i>et al.</i> , 2023); (Rolland and Lyytinen, 2021)	TD in financial services	Partial	×	✓	No quantitative ranking output
(Banker, Liang and Ramasubbu, 2020); (Ramasubbu and Kemerer, 2021); (Rinta-Kahila <i>et al.</i> , 2023)	Enterprise COTS and outsourcing	×	×	✓	No prioritisation framework
(Forsgren, Humble and Kim, 2018); (Nord, Ozkaya and Woody, 2021); (Paudel <i>et al.</i> , 2025)	DORA metrics as TD proxies	✓	×	Partial	No ranking instrument
(Tizio, Armellini and Massacci, 2023); (Markkanen and Frantti, 2023); (Nurse, 2025)	Patch and vulnerability debt	Partial	×	Partial	Not linked to TD prioritisation
(Albarak <i>et al.</i> , 2022); (Khan and Purohit, 2022); (Zytoon, 2020)	TOPSIS / MCDM in IT	×	✓	×	No live data; no trading IT context

Reference	Focus	SL Ind.	Multi-crit.	Fin. IT	Key gap
This thesis	TOPSIS on ITSM/CMDB data	✓	✓	✓	—

The research gap identified in the final row of Table 2.4 defines the problem that Chapter 3 translates into a concrete research design, and that Chapters 4 and 5 address through framework construction and evaluation.

2.8.4 The Research Gap

Related work reviewed in this thesis does not offer a framework that integrates service-level operational indicators, multi-criteria prioritisation and the specific constraints of regulated financial trading IT into a single auditable ranking instrument. Code-centric tools are structurally inapplicable to vendor-hosted platforms, and single-criterion models conflict with both practitioner behaviour and the multidimensional nature of operational risk in trading environments. The resulting framework applies TOPSIS to five operational indicators extracted from live ITSM and CMDB records, producing a ranked prioritisation instrument that operates without source code access and is grounded in data routinely available in regulated enterprise settings. The evaluation follows the same logic used by Greco *et al.* (2019) and Albarak *et al.* (2022): testing whether the rankings hold when weights change, and checking whether the ranking is reproduced under alternative multi-criteria methods and diverges from a single-criterion baseline.

3 Research Design and Methodology

The philosophical approach, research strategy, and design decisions that govern the construction and evaluation of the proposed framework are developed in this chapter. It proceeds in two parts: the first covers the research design from philosophical positioning through to the evaluation approach; the second addresses the ethical and professional considerations arising from the use of operational data in a regulated financial institution.

3.1 Research Design

This study is grounded in a pragmatist research philosophy. Pragmatism holds that the value of a knowledge claim is determined by whether the resulting artefact works in the context for which it was designed (Kelly and Cordeiro, 2020). This position suits the present study because the objective is not to establish universal laws about technical debt, but to construct a quantitative framework useful for prioritisation decisions in a specific enterprise IT environment. Technical debt is treated here as a construct without a single measurement, since tools applied to the same codebase produce substantially different estimates (Amanatidis *et al.*, 2020). The framework therefore grounds its estimates in operationally defined proxies and acknowledges that the resulting scores are representations, not facts.

3.1.1 Research Strategy

Design Science Research (DSR) governs this study, as formalised by Hevner *et al.* (2004). DSR is appropriate here because the primary output is an artefact rather than a theory: a TOPSIS-based prioritisation framework operating on live ITSM

data. The framework is designed to address the gap identified in Chapter 2, where no existing approach integrates service-level operational metrics for technical debt prioritisation in trading IT. Lenarduzzi *et al.* (2021) confirm the broader absence of validated, quantitative prioritisation frameworks across the field; the service-level and enterprise-IT dimension of that gap is established through the synthesis in Chapter 2.

Hevner *et al.* (2004) specify seven guidelines for rigorous DSR, mapped to this study in Table 3.1. On one side sits the problem space: the absence of a service-level TD framework. On the other sits the solution space: the TOPSIS artefact. These are connected through input knowledge, such as operational indicators and MCDM methods, and through research processes, including indicator selection and sensitivity analysis. This structure positions the framework as prescriptive design knowledge applicable to regulated enterprise contexts.

TABLE 3.1: DSR guidelines and their implementation

DSR Guideline	Implementation in this Study
Design as an artefact	The TOPSIS-based prioritisation framework
Problem relevance	Established in Chapter 2: no existing framework integrates service-level operational metrics for TD prioritisation in trading IT
Design evaluation	Sensitivity analysis; convergent validation across alternative methods; discriminant baseline; rank-reversal test
Research contributions	A novel operational indicator set; to the best of current knowledge, the first application of TOPSIS to live ITSM and CMDB data for technical debt prioritisation in financial trading IT
Design as a search process	Iterative refinement of indicator selection (literature candidates audited against ITSM/CMDB availability) and weight assignment
Communication of research	This dissertation; practitioner-accessible scoring output

The process follows Vom Brocke, Hevner and Maedche (2020)'s six-step model: problem identification and motivation (Chapter 1 and Chapter 2), definition of objectives (Chapter 2), design and development of the artefact (Chapter 4), demonstration, evaluation (Chapter 5), and communication (Chapter 6). The sequence is not strictly linear; indicator selection and evaluation criteria were refined iteratively as constraints in the available data emerged during development.

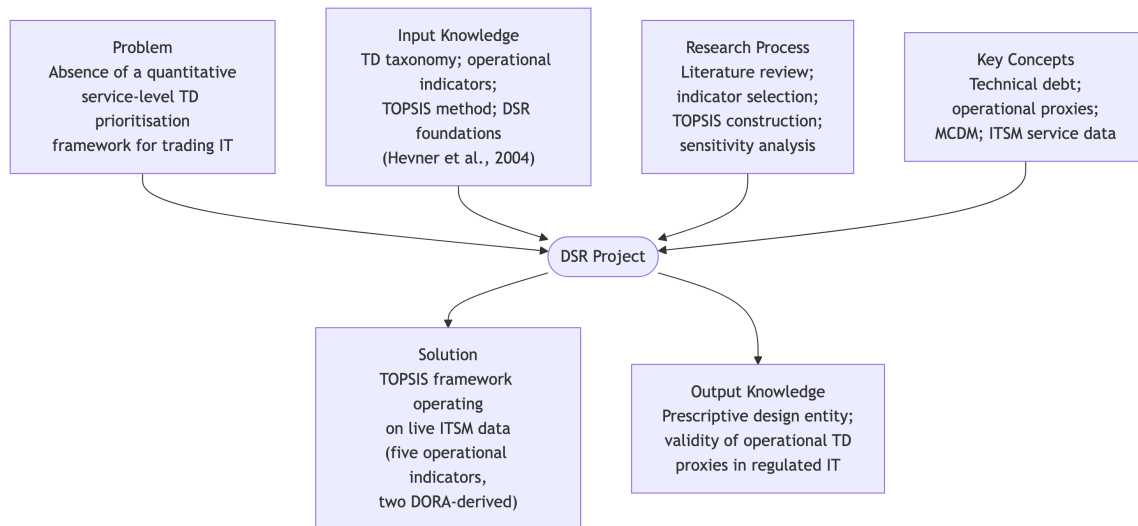


FIGURE 3.1: DSR Grid for the present study, adapted from vom Brocke, Hevner and Maedche (2020)

3.1.2 Research Design

The study is structured as a single-organisation embedded case study, situated within the trading IT division of a systemically important Swiss financial institution. Turnbull, Chugh and Luck (2021) define this design as an investigation into a single instance of observable complex phenomena, constrained by clearly identifiable boundaries. Discrete units of analysis, in this case individual IT services, are examined within that bounded organisational setting.

Three factors make this design appropriate. First, the regulatory, operational, and vendor-platform characteristics specific to Swiss financial IT resist generalisation from software engineering studies conducted elsewhere. Second, access to granular ITSM records is unavailable through surveys or experiments. Third, Hevner *et al.* (2004) specify that a design artefact must be evaluated through rigorous, well-executed methods. The present study satisfies this requirement

by evaluating the framework against operational data from the environment for which it was designed.

The study adopts the individual IT service as the unit of analysis, defined as a distinct configuration item in the organisation's ITSM instance. In this vendor-heavy enterprise environment, source-code access is unavailable for both outsourced components and on-premises vendor software (Ramasubbu and Kemerer, 2021). Service-level operational indicators are therefore tested as candidate proxies for technical debt prioritisation, rather than assumed as established substitutes. The study is cross-sectional, analysing a fixed observation window rather than tracking changes over time, consistent with the objective of demonstrating ranking validity at a point in time. Longitudinal evaluation is identified as a direction for future work.

3.1.3 Data Sources and Collection

The literature review (Chapter 2) identified candidate indicators mapping technical debt dimensions to operational data (see Table 2.3). A data audit against available ITSM and CMDB fields refined this set prior to data collection. Documentation completeness was excluded because no structured ITSM or CMDB field captures it consistently across services; relevant records reside in unstructured repositories not amenable to uniform extraction. Test coverage ratio and architectural coupling were excluded because both require access to assets unavailable in this environment: test coverage is derived from CI/CD build artefacts or source repositories, and architectural coupling is measured through static dependency graph analysis of compiled code (Amanatidis *et al.*, 2020); neither is accessible for vendor-managed nor outsourced components. Raw vulnerability density was excluded as subsumed by patch recency, which provides a more stable CMDB-derived proxy grounded in regulatory patching compliance rather than raw scanner counts whose coverage varies across components.

The five retained indicators are operationalisable from live records held in the organisation's ITSM instance and CMDB. Records were accessed under an agreed arrangement with the Head of Trading IT over a twelve-month observation window (January–December 2025) and anonymised at source, with service

names replaced by alphanumeric identifiers and all team and vendor labels removed before analysis.

TABLE 3.2: Operational definition of indicators used in the TOPSIS framework

Indicator	Operational Definition	Source	Unit	Direction
Incident frequency	Number of incidents per service per year.	ITSM	Incidents / year	Cost
MTTR	Arithmetic mean resolution time for incidents.	ITSM	Hours	Cost
CFR	Weighted incidents divided by the number of changes.	ITSM	Ratio	Cost
Patch recency	Share of service components running within a supported release window.	CMDB	Percent-age (0–100)	Benefit
Unsupported component months	Total months of exposure to unsupported components in the service stack.	CMDB	Months	Cost

The operational definitions above should be read with two caveats. First, incident frequency and MTTR are based on P2, P3 and P4 incidents only, because these are the severity levels used consistently in the case organisation’s ITSM records. Second, CFR is a derived proxy rather than a directly tracked service metric, and is calculated from weighted incidents divided by the number of changes. It approximates the intent of the DORA change failure rate but is not identical to it: a true change failure rate requires deployment-outcome labelling that the case organisation does not record. Because the ratio shares its incident numerator with the incident-frequency indicator, the two criteria are not fully independent. This partial overlap is accepted as a constraint of working with operational data, and its effect on the ordering is bounded by the weight sensitivity analysis in Section 5.2.

Incident frequency, MTTR and CFR proxy the operational consequences of architectural and infrastructure debt. Service instability and slow recovery are

empirically linked to TD severity in vendor-managed portfolios (Ramasubbu and Kemerer, 2021; Nielsen and Skaarup, 2022). Deployment instability is classified as a direct observable symptom of accumulated debt (Forsgren, Humble and Kim, 2018; Nord, Ozkaya and Woody, 2021). Nord, Ozkaya and Woody (2021) further ground the inclusion of MTTR and CFR specifically by classifying delivery tempo degradation as a diagnostic TD signal detectable from production records, independently of source code access. Patch recency captures the proportion of components currently within a vendor- or organisation-supported version window. Unsupported component months measures the cumulative duration of exposure to end-of-life components during the observation period. Together, these two CMDB-derived indicators operationalise infrastructure lifecycle and vulnerability debt following Haki *et al.* (2023), differentiating services with similar current patch status but differing historical debt accumulation. Service-level operational indicators are treated as candidate proxies for technical debt prioritisation rather than established substitutes. The validity of this inference is the central empirical hypothesis tested in Chapter 5.

3.1.4 Framework Design

The analytical core is the Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS), formulated by Hwang and Yoon (1981) and validated through comprehensive reviews of its enterprise applications (Pandey, Komal and Dincer, 2023). TOPSIS addresses the single-criterion prioritisation limitation identified in Chapter 2 by ranking services according to their geometric proximity to ideal solutions after normalisation and weighting. Given a decision matrix with m services and n indicators, the method proceeds as follows.

The matrix is normalised using vector normalisation, which scales each value relative to the column norm to ensure comparability across indicators with different units and ranges (Pandey, Komal and Dincer, 2023):

$$r_{ij} = \frac{x_{ij}}{\sqrt{\sum_{i=1}^m x_{ij}^2}}$$

A weighted normalised matrix is constructed using criterion weight w_j :

$$v_{ij} = w_j \cdot r_{ij}$$

The positive and negative ideal solutions are identified:

$$A^+ = \{v_1^+, \dots, v_n^+\}, \quad A^- = \{v_1^-, \dots, v_n^-\}$$

Euclidean distances from each ideal are calculated:

$$S_i^+ = \sqrt{\sum_{j=1}^n (v_{ij} - v_j^+)^2}, \quad S_i^- = \sqrt{\sum_{j=1}^n (v_{ij} - v_j^-)^2}$$

The relative closeness coefficient C_i constitutes the final score:

$$C_i = \frac{S_i^-}{S_i^+ + S_i^-}, \quad C_i \in [0, 1]$$

A higher C_i indicates that a service is closer to the ideal best operational profile across the full indicator set and therefore has lower technical debt priority. Incident frequency, MTTR, CFR, and unsupported component months are cost criteria where lower values are preferable; patch recency is a benefit criterion where higher values are preferable.

Figure 3.2 illustrates the full pipeline from indicator collection through to ranked output.

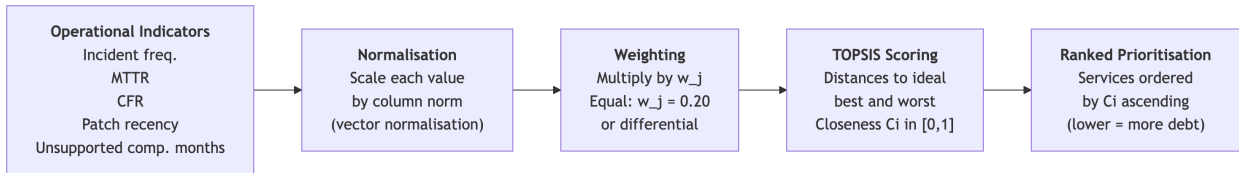


FIGURE 3.2: Framework pipeline: from operational indicators to ranked prioritisation

Two weight configurations are used. A baseline equal-weight configuration ($w_j = 0.20$) serves as the reference case. Equal weighting is the standard default in composite indicator construction in the absence of empirically justified differential weights (Greco *et al.*, 2019), and provides a transparent basis for the

sensitivity analysis that follows. A secondary configuration assigns higher weight to MTTR and patch recency ($w_j = 0.30$ each) and lower weight to the remaining three indicators ($w_j = 0.13333$ each), reflecting a practitioner judgement that service recovery time and omitting to install updates and patches carry greater operational consequence in time-critical trading environments. This weighting is not derived from literature and is treated as an explicit assumption; its influence on rankings is assessed in Chapter 5. The sensitivity of rankings to weight perturbation more broadly is also assessed there, following the practice recommended by Greco *et al.* (2019) for composite indicators.

3.1.5 Evaluation Approach

No external ground-truth measure exists for the non-code debt this framework targets. Code-level tools disagree on the same codebase (Amanatidis *et al.*, 2020). Measurement tooling also remains predominantly code-level, so infrastructure, platform and vulnerability debt are weakly observed. This is the gap established in Chapter 2. The agreed governance scope further precluded interviewing service teams (Section 3.2). The framework is therefore evaluated analytically, in the sense of Hevner *et al.* (2004) Guideline 3. It establishes robustness, method-independence and discriminant value, rather than criterion validity, which is left to future work.

Four strands are applied. The first tests robustness, using Kendall's τ to compare the baseline ranking against weight-perturbed and temporal variants. The second tests convergent validity. Here the same services are re-ranked with Simple Additive Weighting (SAW) and VIKOR, and each ranking's agreement with the TOPSIS order is measured. Such comparison of multi-criteria methods is well established (Shekhovtsov and Saġabun, 2020; Ciardiello and Genovese, 2023). Strong agreement shows that the order reflects the indicator profile, not the mechanics of one method. The third tests discriminant value against a single-criterion ranking by incident frequency. Divergence here shows the framework catches debt a single signal would miss, consistent with Albarak *et al.* (2022). The fourth tests structural rigour. It removes services one at a time and checks whether the order of those remaining holds, addressing the known rank-reversal effect in TOPSIS (García-Cascales and Lamata, 2012; Yang, 2020).

3.2 Ethical and Professional Considerations

The ethical considerations arising from this study concern data confidentiality and legal compliance, the responsible use of prioritisation outputs, and alignment with professional standards.

3.2.1 Data Confidentiality and Anonymisation

The data originate from a live Swiss financial institution that is not identified by name. All service, team, and vendor identifiers were replaced with alphanumeric codes before analysis. No employee data are present: all five indicators are service-level aggregates that cannot be attributed to individuals. The data-sharing arrangement was established with the explicit authorisation of the Head of Trading IT. Only the data fields necessary for the analytical purposes of this study were exported, and no records were retained beyond the defined analytical period. At no point were data transferred outside the organisation's infrastructure.

3.2.2 Legal Compliance

The organisation is subject to the revised Federal Act on Data Protection (nFADP), which entered into force on 1 September 2023 (*Federal Act on Data Protection*, 2023), enforced by the Federal Data Protection and Information Commissioner (FD-PIC). Because the dataset contains no personal data, the nFADP's provisions on personal data processing do not apply directly. The principle of data minimisation, which the nFADP enshrines in alignment with international standards, was nonetheless observed. Only the metrics required for the five TOPSIS indicators were extracted.

As a systemically significant institution, the organisation is also subject to FINMA Circular 2023/1 on operational risks and resilience, in force since 1 January 2024 (*FINMA Circular 2023/1 - Operational risks and resilience – banks*, 2024). This circular requires comprehensive ICT operational risk management, including incident monitoring and service resilience. While the framework benefits from these guidelines, it simultaneously adds value by providing a clearer overview of technical debt. However, the specific metrics used here are organisationally defined

rather than prescribed by FINMA. No ethics committee approval was required under the University of Essex's research ethics framework, and a departmental no-objection confirmation was obtained.

3.2.3 Responsible Use of Prioritisation Outputs

A relative rank assigned to a service carries a risk of being misused to evaluate team performance. High-debt service profiles typically reflect historical design decisions or legacy constraints rather than the competence of the team currently managing the service (Ahmad *et al.*, 2026). This risk is not hypothetical in regulated financial institutions, where priority scores can influence budgets, staffing allocations and post-incident accountability.

The governance scope of this framework was agreed with the Head of Trading IT at the outset of the study. The explicit understanding was that the framework would support a broader, portfolio-level view of service health and inform governance conversations, rather than serve as a definitive or binding prioritisation instrument. The closeness coefficient is a decision-support signal derived from operational proxies, and it was presented as such from the start of the engagement.

Team leads were not involved in the creation of this framework, as the study was conducted solely by the author as part of a master's thesis. No outputs were shared with service teams during the research period. Any operational use of the priority scores beyond the agreed portfolio management scope would require a separate governance decision by the Head of Trading IT before deployment.

3.2.4 Professional Standards and Limitations

This study was conducted in accordance with the BCS Code of Conduct (*Code of Conduct for BCS Members*, 2022), including the obligations to act with integrity, to respect the rights of third parties, and to ensure that the artefact is fit for purpose. Three limitations clarify scope. The single-site design means that indicator weights and TOPSIS configuration are calibrated for one organisation and should not be transferred without revalidation. The cross-sectional window means the framework captures a snapshot; recently remediated services may

be under-ranked. Finally, the five indicators do not exhaust all dimensions of operational technical debt; indicators related to documentation, test coverage, or architectural coupling were excluded due to data availability and remain candidates for future iterations.

4 Framework Development and Results

Twelve months of live ITSM and CMDB records from a regulated Swiss trading institution supply the raw material for this chapter. The indicator values extracted from those records, the normalisation and weighting decisions applied to them, and the TOPSIS scores that result are each documented and justified in the sections below.

4.1 Data Preparation

The trading IT portfolio includes 108 services. Fifty-three of these are small vendor applications such as trading graphical user interfaces and were excluded from the analysis because they require no significant ongoing activity from the portfolio's IT teams.

From the remaining services, six were selected for demonstration on the basis of data completeness. Each holds a continuous twelve-month record across all five indicators in both the ITSM instance and the CMDB, which a substantial share of the wider portfolio does not. All six are registered as active configuration items and have been in operation for more than thirty-six months. Service identifiers were replaced with alphanumeric codes before analysis in line with the anonymisation procedure described in Section 3.2. The dataset contains no team, vendor, or individual identifiers. This is a deliberately small demonstration set, and two consequences follow. First, the services selected tend to carry higher operational activity, so the absolute indicator values are not representative of the full portfolio. Second, a six-service decision matrix limits the discriminatory power of the rank-stability tests, as discussed in Section 5.7. The contribution is

therefore the working instrument and its evaluation logic, not a portfolio-wide ranking, which is identified as future work once indicator coverage across the portfolio is complete.

4.1.1 Indicator Extraction and Quality Checks

The five indicators defined in Section 3.1.3 were extracted from the organisation's ITSM instance and CMDB on 11 June 2026. Only production records were included in the extraction; incidents, changes, and configuration items classified under non-production environments were excluded.

Prior to normalisation, several quality filters were applied to the raw incident data. Incidents sharing a common root cause but recorded as multiple separate tickets, a pattern typical of alarm storms, were consolidated and counted as a single incident to avoid inflating frequency counts. Incidents with an MTTR below 0.1 hours were excluded on the grounds that resolutions at that speed are highly likely to reflect automatic closure following a false alarm or instant recovery rather than genuine remediation effort. To address the upper tail of the MTTR distribution, a 98th percentile cap was applied; incidents exceeding this threshold were excluded as their resolution times are more plausibly attributable to administrative inactivity or elapsed calendar time than to active resolution effort.

All CMDB records were found to be complete. No imputation or worst-case substitution was required.

4.1.2 Data Limitations

The data carry inherent quality limitations that must be acknowledged before interpreting the results. Incident P2/P3/P4 classifications vary across teams, reflecting local triage conventions rather than a uniform severity standard. Patch recency is bounded by CMDB coverage and scanner configuration, so services with incomplete configuration records may appear more current than they are.

Because the organisation does not document CFR directly, it is derived from incident and change records as described in Section 3.1.3. This means incident

frequency and CFR depend on the accuracy and completeness of the incidents that have been opened. Recording practices vary across teams: some contacts arrive by phone or email and whether these result in a formal ITSM record depends on the discipline of the receiving team. Others are handled as defects or bugs within a different issue tracker and never escalated into the ITSM instance. Services where informal handling is common may therefore show lower incident counts and CFR values than their true failure rate warrants, which would systematically compress their priority scores.

These limitations do not invalidate the proxy approach. Scores should be interpreted as ordinal priority signals rather than exact measurements. The sensitivity analysis in Chapter 5 tests whether the ranking is robust to the input variation these data quality factors may introduce.

4.1.3 Operational Indicator Values

Table 4.1 presents the extracted indicator values for all six services. The definitions of these indicators are given in Section 3.1.3.

TABLE 4.1: Extracted indicator values per service (January-December 2025)

Service	Incident Freq. (per year)	MTTR (hrs)	CFR	Patch Recency	Unsup. Comp. Months
SVC-01	64	45.35	0.83	92.45	44
SVC-02	30	25.26	0.19	59.93	1221
SVC-03	21	13.31	0.38	100.00	0
SVC-04	574	41.91	3.38	100.00	0
SVC-05	171	23.72	1.52	100.00	0
SVC-06	302	13.11	3.96	56.25	353

The exact content of the files that have been used for the TOPSIS calculation can be found in the Appendix B.

4.2 Indicator Normalisation and Weighting

Each indicator was normalised using vector normalisation, as specified in Section 3.1.4. This removes differences in scale and unit while preserving proportional relationships between values. A service with an unusually high MTTR retains its full distance from lower-performing services in the normalised space, rather than being compressed toward a fixed upper bound. Min-max normalisation was considered and rejected on this basis. Greco *et al.* (2019) recommend vector normalisation for composite indicators precisely because it does not reduce the discriminatory power of extreme observations.

Two weight configurations are applied, as specified in Section 3.1.4. The baseline equal-weight configuration assigns $w_j = 0.20$ to each of the five criteria and serves as the reference case. The differential configuration assigns higher weights to MTTR and patch recency, consistent with the design rationale established in Chapter 3. The differential weights are $w_{\text{MTTR}} = w_{\text{patch recency}} = 0.30$ and $w_{\text{IncFreq}} = w_{\text{Patch}} = w_{\text{UnsupComp}} \approx 0.13333$.

TABLE 4.2: Weight configurations

Configuration	Incident		Patch		Unsup.
	Freq.	MTTR	CFR	Recency	Comp. Months
Equal (baseline)	0.20000	0.20000	0.20000	0.20000	0.20000
Differential	0.13333	0.30000	0.13333	0.30000	0.13333

4.3 TOPSIS Computation and Rankings

The computation follows the TOPSIS formulation specified in Section 3.1.4 and was executed using the Python artefact in Appendix A. This process calculates the closeness coefficient C_i for each service based on its Euclidean distance from the ideal solution. A higher C_i indicates that a service is closer to the positive ideal solution, while a lower C_i signals higher technical debt priority. Since the framework is intended to guide prioritisation, the service with the least favourable score receives the highest rank.

Table 4.3 presents the full results under both weight configurations. The table is sorted from highest to lowest debt priority, so rank 1 = highest priority for remediation.

TABLE 4.3: TOPSIS closeness coefficients and rankings for 2025.
Higher C_i indicates lower debt priority.

Service	C_i	Rank	C_i	Rank
	(Equal)	(Equal)	(Adjusted)	(Adjusted)
SVC-04	0.4784	1	0.4457	1
SVC-06	0.5095	2	0.5677	3
SVC-02	0.5241	3	0.5238	2
SVC-01	0.7426	4	0.5786	4
SVC-05	0.7792	5	0.7593	5
SVC-03	0.9774	6	0.9808	6

To assess temporal stability, the framework was applied to the 2024 dataset using the same weight configurations. Table 4.4 presents the resulting closeness coefficients and rankings for both configurations.

TABLE 4.4: TOPSIS closeness coefficients and rankings for 2024.
Higher C_i indicates lower debt priority.

Service	C_i	Rank	C_i	Rank
	(Equal)	(Equal)	(Adjusted)	(Adjusted)
SVC-04	0.4838	1	0.5041	1
SVC-06	0.5162	2	0.6044	3
SVC-02	0.7006	3	0.5113	2
SVC-05	0.7059	4	0.7093	5
SVC-01	0.7774	5	0.6251	4
SVC-03	0.8492	6	0.8816	6

The change in SVC-02 between the two years is substantial: cumulative unsupported-component exposure rises from zero to 1,221 component-months.

SVC-02 is built from more than two hundred components, and roughly half of these passed their vendor end-of-support date during 2025, derived from the same complete CMDB extract that underlies the unsupported-component-months indicator (Appendix B).

The calculations presented in this chapter constitute the analytical output of the framework. The closeness coefficients and rankings produced under both weight configurations, for 2025 and for temporal comparison for 2024, are carried forward as the empirical basis for the evaluation that follows. Chapter 5 assesses whether these rankings are stable under weight variation and across observation periods, tests their stability under weight and temporal variation, their agreement with alternative multi-criteria methods, and their divergence from a single-criterion baseline. Finally, it reflects on the implications for technical debt prioritisation within the organisation's trading IT portfolio.

5 Evaluation and Discussion

The framework developed in Chapter 4 ranks six production services from five operational indicators held in the organisation's ITSM instance and CMDB. A ranking only earns trust once it behaves predictably, so this chapter tests the order before interpreting it: whether it is stable when criterion weights and observation years vary, whether alternative multi-criteria methods reproduce it, whether it diverges from a single-criterion view, and whether it survives the removal of individual services. A final section reads the rankings for their practical meaning, each test interpreted against the research questions.

5.1 Evaluation Criteria

The framework is evaluated against the criteria set out in Section 3.1.5, which together form an analytical evaluation in the sense of Hevner *et al.* (2004) Guideline 3. Five properties are at issue. Ranking stability means the priority order is insensitive to reasonable weight variation and holds across observation periods, assessed with Kendall's tau across the equal-weight and adjusted-weight rankings for 2025 and 2024. Convergent validity means alternative multi-criteria methods reproduce the ordering; discriminant value means it departs from a single-criterion incident-frequency baseline. Structural rigour means the order of the remaining services holds when any one is removed, and practical validity means the rankings can be read in terms of the service profiles they describe.

5.2 Stability Results

Kendall's tau (τ) measures the similarity between two rankings and is well established in MCDA evaluation (Shekhovtsov, 2021). For every pair of services it counts the concordant pairs, subtracts the discordant ones and scales the result

to $[-1, 1]$, where $\tau = 1$ is perfect agreement and $\tau = -1$ a complete reversal. It is a dependable choice here: Shekhovtsov (2021) show through exhaustive permutation analysis that it tracks Spearman's and Weighted Spearman's coefficients closely ($r = 0.982$ and $r = 0.970$ respectively).

With only six services, the p-value is obtained exactly: all $6! = 720$ orderings are enumerated, and the p-value is the share producing a $|\tau|$ at least as large as the one observed. This small sample limits the test's power, but the limitation comes from the size of the portfolio rather than weak agreement between rankings, since p-values fall monotonically as the sample grows (Chén *et al.*, 2023). A value near the conventional 0.05 threshold should therefore not be read as a clean verdict of independence; as Chén *et al.* (2023) note, borderline cases call for contextual judgement rather than mechanical rejection.

Table 5.1 reports the concordance values for all four comparisons.

TABLE 5.1: Kendall's tau concordance values across all ranking comparisons.

Comparison	τ	p
Weight sensitivity 2025 (equal vs adjusted)	0.867	0.017
Weight sensitivity 2024 (equal vs adjusted)	0.733	0.056
Temporal stability equal weights (2025 vs 2024)	0.867	0.017
Temporal stability adjusted weights (2025 vs 2024)	1.000	0.003

Given the small sample, these p-values are reported for transparency and read as effect sizes rather than accept-or-reject decisions, in line with the caution above.

The 2025 weight sensitivity result ($\tau = 0.867$, $p = 0.017$) shows strong agreement between the two configurations. The only movement is a swap between SVC-06 and SVC-02 at ranks 2 and 3: under adjusted weights, SVC-02's very high

unsupported component months and weaker patch recency pull it above SVC-06. The top and bottom of the ranking are identical under both.

Technical Debt Priority

Weights: 0.20 per criterion | Red = Rank 1 (highest remediation priority)

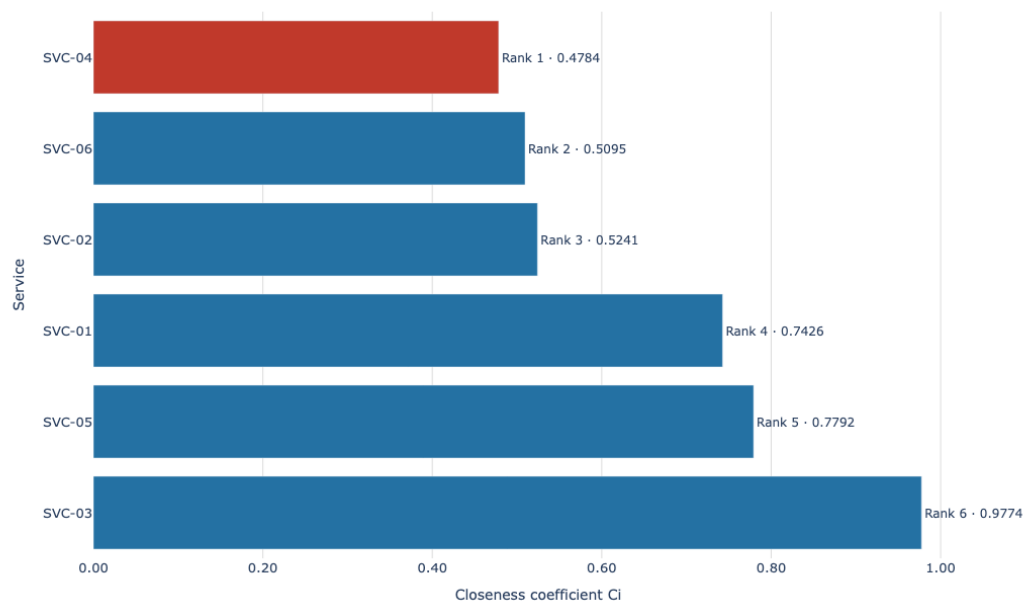


FIGURE 5.1: Technical debt priority ranking for 2025, equal weights.

Technical Debt Priority

Weights: 0.3 MTTR and Patch recency, rest 0.13333 | Red = Rank 1 (highest remediation priority)

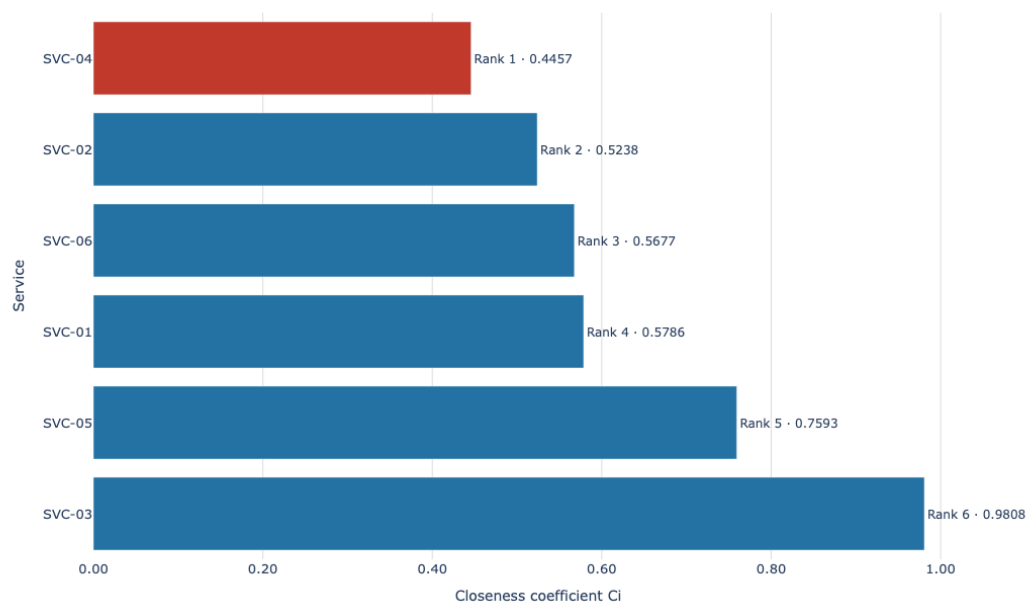


FIGURE 5.2: Technical debt priority ranking for 2025, adjusted weights.

Temporal stability under the adjusted configuration is perfect ($\tau = 1.000, p = 0.003$): the ordering is identical across both years. Under equal weights it stays strong ($\tau = 0.867, p = 0.017$), with only SVC-01 and SVC-05 trading ranks 4 and 5. That reflects a real year-on-year shift in their incident, recovery and patch profiles rather than instability in the model.

The 2024 weight sensitivity result ($\tau = 0.733, p = 0.056$) is the weakest in the set. Two pairs swap rather than the single swap seen in 2025, because the middle-ranking services have more tightly grouped scores in 2024, making their order more sensitive to a change in weights. This is not a structural weakness; it shows that differential weighting matters more when services carry similar profiles.

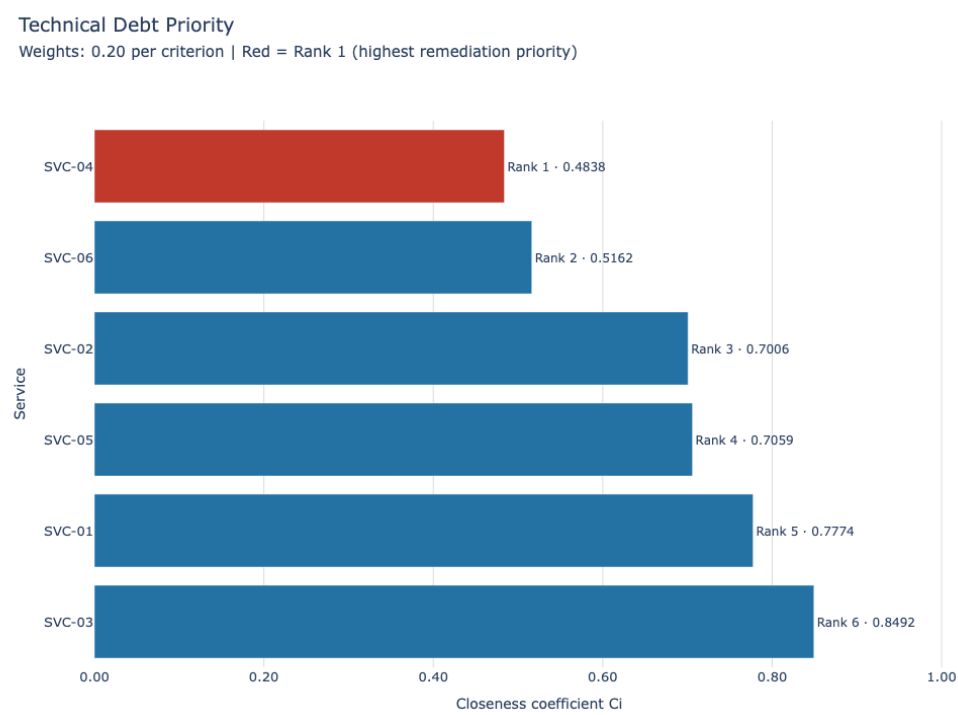


FIGURE 5.3: Technical debt priority ranking for 2024, equal weights.

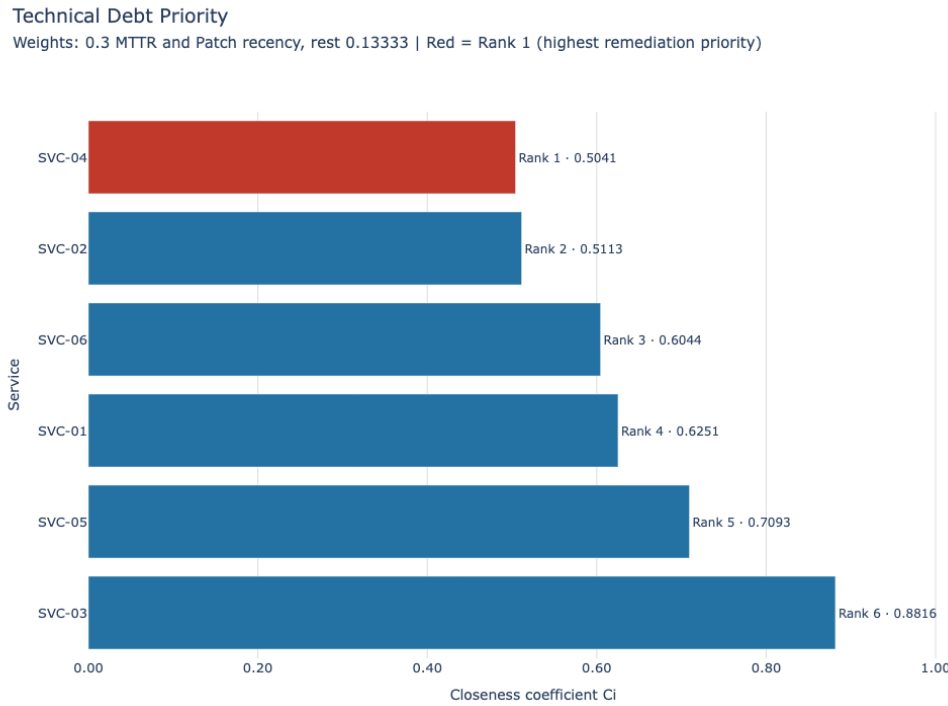


FIGURE 5.4: Technical debt priority ranking for 2024, adjusted weights.

Taken together, these results address RQ3 directly: the rankings are robust to reasonable variation in criterion weights and consistent across observation years, with the top and bottom positions stable in every run.

5.3 Convergent Validity Across Methods

To test convergent validity, the six services were re-ranked with SAW and VIKOR and each ordering compared against the TOPSIS result. Because the dataset contains zero values, SAW used Weitendorf (min-max) normalisation, which avoids the division by zero that classic cost normalisation would produce (Thien, Dung and Trung, 2024). VIKOR followed the standard S, R and Q procedure with $v = 0.5$. Table 5.2 reports agreement across both configurations and years; Table 5.3 gives the ranks each method assigns in the 2025 equal-weight reference case.

Agreement is moderate rather than exact, with τ from 0.600 to 0.867, and the pattern behind those figures is consistent. All three methods place the same three services in the high-priority half and the same three in the low-priority

half, and SVC-03 comes last under every method, configuration and year. They part company only on the internal order of the high-priority cluster: TOPSIS ranks SVC-04 first, on its very high incident volume and CFR, whereas SAW and VIKOR both promote SVC-06, whose worst-in-portfolio patch recency and high unsupported-component exposure give it the largest single-criterion regret. The broad prioritisation is therefore largely independent of method, even though the single highest-priority service is not. That matters when the framework informs an individual remediation choice rather than portfolio-level triage, a limitation revisited in Section 5.7.

TABLE 5.2: Rank agreement (Kendall’s τ) between TOPSIS and the two alternative methods.

Configuration	TOPSIS vs SAW	TOPSIS vs VIKOR
2025, equal weights	0.867	0.867
2025, adjusted weights	0.600	0.600
2024, equal weights	0.733	0.867
2024, adjusted weights	0.733	0.600

TABLE 5.3: Debt-priority rank by method, 2025 equal weights.

Service	TOPSIS	SAW	VIKOR
SVC-04	1	2	2
SVC-06	2	1	1
SVC-02	3	3	3
SVC-01	4	4	4
SVC-05	5	5	5
SVC-03	6	6	6

5.4 Discriminant Value Against a Single-Criterion Baseline

Discriminant value was tested against the simplest single-signal alternative: a ranking by incident frequency alone. The two orderings diverge, with Kendall’s τ

of 0.600 in 2025 and 0.200 in 2024, and Table 5.4 sets out the 2025 comparison. The divergence concentrates in two services. SVC-02 rises against the incident-only view because its infrastructure-lifecycle debt, carried in unsupported-component exposure and low patch recency, is invisible to an incident count, while SVC-05 falls because its high incident volume is offset by full patch recency and no unsupported components. This is the value the multi-criteria design adds over a single operational signal, consistent with Albarak *et al.* (2022), who found that multi-criteria prioritisation surfaces items single-factor approaches defer. The wider divergence in 2024 is the same mechanism operating more strongly: services carrying infrastructure or recovery-time debt, but only modest incident counts rank far higher under the framework than under incident volume alone.

TABLE 5.4: TOPSIS rank against an incident-frequency ranking, 2025 equal weights.

Service	TOPSIS rank	Incident-only	
		rank	Shift
SVC-04	1	1	0
SVC-06	2	2	0
SVC-02	3	5	+2
SVC-01	4	4	0
SVC-05	5	3	-2
SVC-03	6	6	0

5.5 Rank Reversal

Structural rigour was examined with a rank-reversal test: each service was removed in turn and the remainder re-ranked. Table 5.5 reports the 2025 equal-weight results. The order holds in three of the six removals. In the other three, a single adjacent pair changes places: removing SVC-02 swaps the two highest-priority services, SVC-04 and SVC-06, while removing either SVC-04 or SVC-06 swaps the middle pair, SVC-01 and SVC-05. Each affected pair sits within a narrow closeness-coefficient margin, and no service crosses between the high-priority and low-priority halves under any removal; the lowest-priority service never moves. Rank reversal is therefore present but confined to fine ordering within

already adjacent clusters, the familiar limitation of distance-based aggregation identified by García-Cascales and Lamata (2012). Where a fixed reference frame is required, the normalisation proposed by Yang (2020) removes the effect.

TABLE 5.5: Effect of removing each service on the order of the remaining five, 2025 equal weights.

Service removed	Outcome
SVC-01	Order preserved
SVC-02	SVC-04 and SVC-06 exchange ranks 1 and 2
SVC-03	Order preserved
SVC-04	SVC-01 and SVC-05 exchange ranks
SVC-05	Order preserved
SVC-06	SVC-01 and SVC-05 exchange ranks

5.6 Practical Validity

SVC-04 holds rank 1 in all four calculations, the highest technical debt priority in the portfolio. Its 2025 profile pairs the highest incident frequency, 574 per year, with an MTTR of 41.91 hours and a CFR of 3.38, yet carries no unsupported components and full patch recency. The high incident volume and change failure rate point to debt that is operational rather than infrastructural, consistent with the architectural debt symptoms described in the literature (Ramasubbu and Kemerer, 2021; de Toledo *et al.*, 2021).

At the other end, SVC-03 has the lowest debt priority in every run: the fewest incidents, a low MTTR, a low CFR, full patch recency and no unsupported components. This profile anchors the negative ideal solution and offers a concrete reference for a well-maintained service.

The most revealing service is SVC-02. It ranks third under equal weights and rises to second under adjusted weights, driven by an unsupported component months value of 1,221, by far the largest in the portfolio, together with the lowest patch recency score. This is exactly the signal the framework is built to surface: a service that looks operationally stable in isolation but carries structural debt

only multi-criteria analysis brings into view, consistent with the infrastructure lifecycle debt category described by Haki *et al.* (2023).

These results address RQ2. A strict causal claim cannot be drawn from cross-sectional data alone, but the rankings match the operational debt profiles the literature would predict for each position (Ramasubbu and Kemerer, 2021; Nord, Ozkaya and Woody, 2021; Haki *et al.*, 2023). The framework yields an ordinal priority signal that can be read against the debt dimensions mapped in Table 2.3.

5.7 Limitations of the Evaluation

The most fundamental limitation is that every strand of the evaluation draws on the same dataset and indicator set used to build the framework. The triangulation and baseline comparisons establish convergent and discriminant validity, but not criterion validity against an independent measure of technical debt, because no such measure exists for the non-code debt in scope. External validation is therefore the main avenue for future work.

The small portfolio of six services limits the power of the Kendall's tau tests. A concordance of $\tau = 0.733$ cannot reach significance at the conventional threshold when only 720 permutations are possible, even though the agreement it represents is substantively meaningful. A larger service set would ease this constraint.

The cross-sectional design captures only a snapshot. A service that has recently completed significant remediation may look worse than its current state warrants, while one whose debt is still growing may not yet show the full consequences within a single annual window. Longitudinal evaluation is flagged as a priority for future work in Chapter 6.

Finally, the data quality limitations set out in Section 4.1.2 apply here too. Inconsistent incident recording, partial issue capture split across the organisation's ITSM platform and its issue tracker, and the dependence of CFR on documentation discipline all add noise to the input. The stability results offset this in part, but middle-position rankings still warrant careful reading.

6 Conclusion

6.1 Research Summary

6.2 Findings Against Research Questions

6.3 Contribution to Knowledge

6.4 Practical Contribution

6.5 Limitations and Future Work

6.6 Reflective Note

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A Appendix A: Python Code

The content in this appendix is the exact artefact which has been used to calculate the TOPSIS rankings. Further artefacts which have been used to do other calculations can be found on the authors GitHub Repository: <https://github.com/TobiZeier/MScProject>.

```
#!/usr/bin/env python
"""
TOPSIS scoring for technical debt prioritisation.
Reads a CSV of services, scores them, prints a ranked table and
↳ saves a bar chart.
"""

import argparse
import sys

import numpy as np
import pandas as pd
import plotly.graph_objects as go

CRITERIA = [
    "incident_freq",
    "mttr_hrs",
    "cfr",
    "patch_recency",
    "unsupported_months",
]
```

```

# patch_recency is a benefit criterion - higher = more recent =
↳ better
# everything else - lower = better
COST_CRITERIA = [
    "incident_freq",
    "mttr_hrs",
    "cfr",
    "unsupported_months",
]

WEIGHTS = np.array([0.2, 0.2, 0.2, 0.2, 0.2], dtype=float)
#WEIGHTS = np.array([0.13333, 0.3, 0.3, 0.13333, 0.13333],
↳ dtype=float)

OUTPUT_CHART = "topsis_ranking.png"

def run_topsis(df, criteria, cost_criteria, weights):
    if not np.isclose(weights.sum(), 1.0):
        raise ValueError(f"Weights should sum to 1.0, got
↳ {weights.sum():.4f}")

    X = df[criteria].to_numpy(dtype=float)

    # normalise each column
    norms = np.sqrt((X ** 2).sum(axis=0))
    if np.any(norms == 0):
        raise ValueError("One or more criterion columns are all
↳ zeros - normalisation not possible.")
    R = X / norms
    V = R * weights

    # figure out ideal best/worst per criterion

```

```

benefit = np.array([c not in cost_criteria for c in criteria])
ideal_best = np.where(benefit, V.max(axis=0), V.min(axis=0))
ideal_worst = np.where(benefit, V.min(axis=0), V.max(axis=0))

d_best = np.sqrt(((V - ideal_best) ** 2).sum(axis=1))
d_worst = np.sqrt(((V - ideal_worst) ** 2).sum(axis=1))

total = d_best + d_worst
if np.any(total == 0):
    raise ValueError("A row is equidistant from both ideals -
        ↪ closeness coefficient is undefined.")

# Ci close to 0 = closest to worst (highest debt priority)
Ci = d_worst / total
#print(Ci)

out = df[["service"]].copy()
out["closeness_coeff"] = Ci
out["rank"] = out["closeness_coeff"].rank(ascending=True,
    ↪ method="min").astype(int)
return out.sort_values(["rank", "closeness_coeff",
    ↪ "service"]).reset_index(drop=True)

def load_data(path):
    try:
        df = pd.read_csv(path)
    except FileNotFoundError:
        sys.exit(f"Couldn't find file: {path}")

    needed = {"service"} | set(CRITERIA)
    missing = needed - set(df.columns)
    if missing:

```

```

        sys.exit(f"CSV is missing these columns:
↳ {sorted(missing)}")

for col in CRITERIA:
    df[col] = pd.to_numeric(df[col], errors="coerce")

bad = [c for c in CRITERIA if df[c].isnull().any()]
if bad:
    sys.exit(f"Non-numeric or missing values found in: {bad}")

if df["service"].isnull().any():
    sys.exit("Some rows in the service column are empty.")

return df

def print_results(result):
    print()
    print("=" * 56)
    print(" TOPSIS Results - Equal Weights (0.20 per criterion)")
    print("=" * 56)
    print(f"   {'Service':<12} {'Closeness Coeff (Ci)':>22}
↳   {'Rank':>6}")
    print("-" * 56)
    for _, row in result.iterrows():
        print(f"   {row['service']:<12}
↳   {row['closeness_coeff']:>22.4f} {row['rank']:>6}")
    print("=" * 56)
    print(" Rank 1 = highest TD priority | Ci near 0 = worst")
    print(f" Rank {len(result)} = lowest TD priority | Ci near
↳ 1 = best")
    print()

```

```

def save_chart(result, output_path):
    df_plot = result.sort_values(["rank", "closeness_coeff"],
    ↪ ascending=[True, False]).reset_index(drop=True)
    df_plot["label"] = df_plot.apply(lambda r: f"Rank
    ↪ {int(r['rank'])} · {r['closeness_coeff']:.4f}", axis=1)

    colors = ["#c0392b" if r == 1 else "#2471a3" for r in
    ↪ df_plot["rank"]]

    fig = go.Figure(go.Bar(
        y=df_plot["service"],
        x=df_plot["closeness_coeff"],
        orientation="h",
        marker_color=colors,
        text=df_plot["label"],
        textposition="outside",
        cliponaxis=False,
        insidetextanchor="middle",
    ))

    fig.update_layout(
        width=1100,
        height=max(420, 95 * len(df_plot) + 170),
        title=dict(
            text=(
                "Technical Debt Priority<br>"
                "<span style='font-size:15px;font-weight:normal;'>"
                "Weights: 0.20 per criterion | "
                "Red = Rank 1 (highest remediation priority)"
                "</span>"
            )
        )
    )

```

```
        ),
        font=dict(size=19),
        y=0.93,
    ),
    plot_bgcolor="white",
    paper_bgcolor="white",
    margin=dict(t=140, b=60, l=130, r=90),
    xaxis=dict(
        title="Closeness coefficient Ci",
        range=[0, 1.05],
        tickformat=".2f",
        gridcolor="#dddddd",
        zeroline=False,
        tickfont=dict(size=13),
    ),
    yaxis=dict(
        title="Service",
        autorange="reversed",
        tickfont=dict(size=13),
    ),
    showlegend=False,
    uniformtext_minsize=11,
    uniformtext_mode="hide",
)

try:
    fig.write_image(output_path)
except Exception as exc:
    raise RuntimeError(
        "Chart export failed"
    ) from exc

print(f"Chart saved: {output_path}")
```

```

def _run_tests():
    # sanity check: A is ideal, C is worst, B sits in the middle
    test = pd.DataFrame({
        "service": ["A", "B", "C"],
        "cost1": [1.4, 2.6, 4.9],
        "benefit1": [5.2, 3.1, 0.8],
    })
    w = np.array([0.5, 0.5])
    cr = ["cost1", "benefit1"]
    cc = ["cost1"]

    res = run_topsis(test, cr, cc, w)
    ci = res.set_index("service")["closeness_coeff"]

    assert abs(ci["A"] - 1.0) < 1e-6, f"Expected Ci(A) 1.0, got {ci['A']}"
    assert abs(ci["C"] - 0.0) < 1e-6, f"Expected Ci(C) 0.0, got {ci['C']}"
    assert 0.0 < ci["B"] < 1.0, f"Expected 0 < Ci(B) < 1, got {ci['B']}"
    assert res.loc[res["service"] == "C", "rank"].values[0] == 1
    assert res.loc[res["service"] == "A", "rank"].values[0] == 3

def main():
    parser = argparse.ArgumentParser()
    parser.add_argument("csv_file", help="Path to input CSV file")
    args = parser.parse_args()

    #_run_tests()
    df = load_data(args.csv_file)

```

```
    result = run_topsis(df, CRITERIA, COST_CRITERIA, WEIGHTS)
    print_results(result)
    save_chart(result, OUTPUT_CHART)

if __name__ == "__main__":
    main()
```

B Appendix B: Data

This appendix contains the contents of the CSV files that were used as input for the TOPSIS calculations presented in Chapter 4.

Filename: services2025.csv Filesize: 253 bytes

```
service,incident_freq,mtrr_hrs,cfr,patch_recency,unsupported_months
SVC-01,64,45.35,0.83,92.45,44
SVC-02,30,25.26,0.19,59.93,1221
SVC-03,21,13.31,0.38,100.00,0
SVC-04,574,41.91,3.38,100.00,0
SVC-05,171,23.72,1.52,100.00,0
SVC-06,302,13.11,3.96,56.25,353
```

Filename: services2024.csv Filesize: 250 bytes

```
service,incident_freq,mtrr_hrs,cfr,patch_recency,unsupported_months
SVC-01,65,55.37,0.29,100.00,0
SVC-02,39,81.35,0.11,100.00,0
SVC-03,63,9.77,0.86,100.00,0
SVC-04,410,41.53,2.19,100.00,0
SVC-05,140,30.30,1.41,100.00,0
SVC-06,43,15.87,0.86,81.25,329
```

C Appendix C: Full Evaluation Results

This appendix records the complete output behind the summary tables in Chapter 5. It lists, for every weight configuration and both observation years, the per-service scores produced by each method, together with the full convergent, discriminant, rank-reversal and temporal results. In the score tables, C_i is the TOPSIS closeness coefficient, SAW is the Simple Additive Weighting score, and S , R and Q are the VIKOR utility, regret and compromise measures. The final column gives each service's rank under TOPSIS, SAW and VIKOR in that order, where rank 1 is the highest debt priority. Rows are ordered by TOPSIS rank.

C.1 Per-Service Scores and Rankings

TABLE C.1: Full scores and rankings, 2025 equal weights.

Service	C_i	SAW	S	R	Q	Rank (T, S, V)
SVC-04	0.4784	0.4521	0.5479	0.2000	0.9895	1, 2, 2
SVC-06	0.5095	0.4406	0.5594	0.2000	1.0000	2, 1, 1
SVC-02	0.5241	0.5382	0.4618	0.2000	0.9109	3, 3, 3
SVC-01	0.7426	0.7088	0.2912	0.2000	0.7553	4, 4, 4
SVC-05	0.7792	0.8094	0.1906	0.0706	0.3228	5, 5, 5
SVC-03	0.9774	0.9887	0.0113	0.0101	0.0000	6, 6, 6

TABLE C.2: Full scores and rankings, 2025 adjusted weights.

Service	C_i	SAW	S	R	Q	Rank (T, S, V)
SVC-04	0.4457	0.4859	0.5141	0.2680	0.9214	1, 3, 3

Service	C_i	SAW	S	R	Q	Rank (T, S, V)
SVC-02	0.5238	0.4767	0.5233	0.2748	0.9416	2, 2, 2
SVC-06	0.5677	0.4604	0.5396	0.3000	1.0000	3, 1, 1
SVC-01	0.5786	0.6104	0.3896	0.3000	0.8587	4, 4, 4
SVC-05	0.7593	0.8181	0.1819	0.0987	0.3201	5, 5, 5
SVC-03	0.9808	0.9914	0.0086	0.0067	0.0000	6, 6, 6

TABLE C.3: Full scores and rankings, 2024 equal weights.

Service	C_i	SAW	S	R	Q	Rank (T, S, V)
SVC-04	0.4838	0.5113	0.4887	0.2000	0.9968	1, 2, 2
SVC-06	0.5162	0.5087	0.4913	0.2000	1.0000	2, 1, 1
SVC-02	0.7006	0.8000	0.2000	0.2000	0.6415	3, 4, 3
SVC-05	0.7059	0.7632	0.2368	0.1250	0.3935	4, 3, 4
SVC-01	0.7774	0.8413	0.1587	0.1274	0.3069	5, 5, 5
SVC-03	0.8492	0.9149	0.0851	0.0721	0.0000	6, 6, 6

TABLE C.4: Full scores and rankings, 2024 adjusted weights.

Service	C_i	SAW	S	R	Q	Rank (T, S, V)
SVC-04	0.5041	0.6002	0.3998	0.1333	0.5490	1, 2, 3
SVC-02	0.5113	0.7000	0.3000	0.3000	0.7693	2, 3, 2
SVC-06	0.6044	0.4916	0.5084	0.3000	1.0000	3, 1, 1
SVC-01	0.6251	0.7880	0.2120	0.1911	0.4558	4, 4, 4
SVC-05	0.7093	0.7943	0.2057	0.0860	0.2403	5, 5, 5
SVC-03	0.8816	0.9433	0.0567	0.0481	0.0000	6, 6, 6

C.2 Convergent Agreement

TABLE C.5: Rank agreement (Kendall's τ) between TOPSIS and the alternative methods.

Configuration	TOPSIS vs SAW	TOPSIS vs VIKOR
2025, equal	0.867	0.867
2025, adjusted	0.600	0.600
2024, equal	0.733	0.867
2024, adjusted	0.733	0.600

C.3 Discriminant Comparison

TABLE C.6: TOPSIS ranking against a ranking by incident frequency alone, equal weights.

Year	TOPSIS order	Incident-only order	τ
2025	SVC-04, SVC-06,	SVC-04, SVC-06,	0.600
	SVC-02, SVC-01,	SVC-05, SVC-01,	
	SVC-05, SVC-03	SVC-02, SVC-03	
2024	SVC-04, SVC-06,	SVC-04, SVC-05,	0.200
	SVC-02, SVC-05,	SVC-01, SVC-03,	
	SVC-01, SVC-03	SVC-06, SVC-02	

C.4 Rank-Reversal Detail

TABLE C.7: Order of the remaining five services after each removal, equal weights. Entries reading preserved leave the reference order unchanged.

Service removed	2025 outcome	2024 outcome
SVC-01	preserved	preserved
SVC-02	SVC-06, SVC-04, SVC-01, SVC-05, SVC-03	preserved
SVC-03	preserved	SVC-04, SVC-06, SVC-05, SVC-02, SVC-01

Service removed	2025 outcome	2024 outcome
SVC-04	SVC-06, SVC-02, SVC-05, SVC-01, SVC-03	SVC-06, SVC-05, SVC-02, SVC-01, SVC-03
SVC-05	preserved	preserved
SVC-06	SVC-04, SVC-02, SVC-05, SVC-01, SVC-03	SVC-04, SVC-05, SVC-02, SVC-01, SVC-03

C.5 Temporal Stability

TABLE C.8: TOPSIS rankings across the two observation years.

Configuration	2025 order	2024 order	τ
Equal	SVC-04, SVC-06, SVC-02, SVC-01, SVC-05, SVC-03	SVC-04, SVC-06, SVC-02, SVC-05, SVC-01, SVC-03	0.867
Adjusted	SVC-04, SVC-02, SVC-06, SVC-01, SVC-05, SVC-03	SVC-04, SVC-02, SVC-06, SVC-01, SVC-05, SVC-03	1.000